SUGAMMADEX

Reverses rocuronium/vecuronium

Reduces hormonal contraceptive effect

CHEMICAL NAME Octakis-(6-deoxy-6-S-(2-carboxyethyl)-6-thio)-γ-cyclodextrin, octasodium salt

CLASS Selective relaxant binding agent (modified γ-cyclodextrin)

TRADE NAMES Bridion

THERAPEUTIC USES / INDICATIONS

- Routine reversal of neuromuscular block induced by rocuronium or vecuronium.
- Immediate reversal after rocuronium in "cannot intubate, cannot ventilate" situations.

ADMINISTRATION

- **IV** single bolus over 10 seconds into running line.
- No anticholinergic needed.
- Dose based on actual body weight and depth of block (TOF/PTC).

DOSAGES

- **Moderate block (TOF ≥2 twitches):** 2 mg/kg.
- **Deep block (PTC 1–2):** 4 mg/kg.
- **Immediate reversal after 1.2 mg/kg rocuronium:** 16 mg/kg.
- Children (2–17 yr): 2 mg/kg for routine reversal.

PHARMACOKINETICS

- **Onset:** reversal within 2–3 min (even from deep block).
- Not metabolised; sugammadex–rocuronium complex excreted unchanged in urine.
- **t½:** 2 h; not recommended if CrCl <30 mL/min.

PHARMACODYNAMICS

- Encapsulates **aminosteroid** blockers in plasma → removes them from neuromuscular junction.
- No effect on acetylcholinesterase → no muscarinic side effects.
- Ineffective against suxamethonium and benzylisoquinoliniums (atracurium).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Dysgeusia (metallic taste).
- Cough, movement, grimacing (light anaesthesia).
- Nausea.

ADVERSE EFFECTS (SERIOUS)

- Anaphylaxis and hypersensitivity.
- Marked bradycardia, cardiac arrest (rare).
- Prolonged coagulation times (transient).
- Bronchospasm.

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity to sugammadex.
- **Caution:** severe renal impairment, coagulopathy.

KEY INTERACTIONS

- Hormonal contraceptives: one dose equals one missed pill — use extra contraception for 7 days.
- **Toremifene, fusidic acid:** may displace rocuronium → delayed recovery.

NURSING MANAGEMENT

- Confirm blocker was rocuronium or vecuronium.
- Monitor HR, BP, SpO_2, breathing after reversal.
- Observe for allergic reaction for 30 min.
- Assess for re-curarisation (weakness) in recovery.
- Inform women of contraception interaction.

PATIENT EDUCATION

- If you use the pill, implant or vaginal ring, use condoms for the next 7 days.
- Report any rash or breathing difficulty.

KEY POINTS

- Encapsulates rocuronium/vecuronium.
- No anticholinergic needed.
- Reverses even deep block within minutes.
- Reduces hormonal contraceptive efficacy for 7 days.

SOURCES CHECKED **[99]** Sugammadex 100 mg/ mL solution for injection (UK SmPC) **[100]** Sugammadex, IV (US label)
[101] Sugammadex, IV (US label, Hikma Pharmaceuticals USA Inc)

Full references p. 432–457